SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I A.G.Jayapradhan (192471016) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

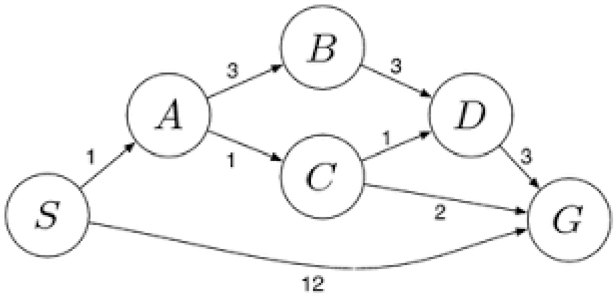
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem (4-gallon & 3-gallon jugs)
Initial State
(4-gallon, 3-gallon) = **(0,0)**
Steps

Step	Action	State (4G,3G)
1	Fill 3-gallon jug	(0,3)
2	Pour 3G into 4G	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4G until full (only 1 gallon needed)	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons from 3G into 4G	**(2,0)**

Final Answer
The **4-gallon jug contains exactly 2 gallons.**

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2. Mars Rover Agent
(i) What are the percepts for this agent?
Percepts are the information collected by the rover through its sensors.

- Terrain
- Rocks
- Soil samples
- Obstacles
- Temperature
- Images
- Distance
- Battery level
- GPS/Location

(ii) Characterize the operating environment

- **Partially Observable** – cannot observe everything.
- **Sequential** – current action affects future actions.
- **Dynamic** – environment changes.
- **Stochastic** – uncertain outcomes.
- **Continuous** – movement and time are continuous.
- **Single Agent** – only the rover performs tasks.

(iii) Actions the agent can take

- Move forward/backward
- Turn left/right
- Capture images
- Collect soil samples
- Drill rocks
- Send data to Earth
- Avoid obstacles
- Recharge (solar panels)

(iv) Performance Measure
The rover performs well if it

- Collects maximum useful samples

- Avoids collisions
- Uses minimum energy
- Completes mission quickly
- Sends accurate scientific data

(v) Suitable Agent Architecture
Goal-Based Agent (or Utility-Based Agent)
Reason:

- Works toward mission goals.
- Chooses actions that maximize mission success.
- Can plan paths and avoid obstacles intelligently.

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3. 8-Queens Problem
Problem Formulation
Initial State
Empty 8×8 chessboard.
State Space
All possible arrangements of 8 queens.
Actions
Place one queen in a safe position in the next column.
Goal Test

- Eight queens are placed.
- No two queens attack each other.

Path Cost
Each placement costs 1.

Components

Start
↓
Empty Board
↓
Place Queen in Column 1
↓
Check Safe Position
↓
Place Queen in Column 2
↓
Continue Until Column 8
↓
No Conflicts?
↓
Yes
↓
Goal State

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4. OLA Cab Booking Problem
Suitable Problem-Solving Agent
Utility-Based Agent
Why?
The customer chooses a cab based on

- Cost
- Time
- Comfort
- Cab availability
- Travel distance

The utility-based agent selects the option with the highest overall satisfaction.

Pseudocode

```
START

Input source and destination

Get available cabs

For each cab
   Calculate fare
   Calculate arrival time
   Calculate travel time
   Calculate comfort score
   Calculate utility score
End For

Select cab with maximum utility

Display selected cab

END
```

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5. Uniform Cost Search (UCS)

Graph Costs

Edges:

- S → A = 1
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3
- S → G = 12

UCS Execution

Step 1

Start at **S**

Frontier

A(1)

G(12)

Step 2

Expand **A(1)**

C = 2

B = 4
G = 12

Step 3
Expand **C(2)**
D = 3
G = 4
B = 4

Step 4
Expand **D(3)**
G = 6
B = 4
Existing path to G is already **4**, so keep **4**.

Step 5
Expand **G(4)**
Goal reached.

Shortest Path
S → A → C → G
Total Cost
1 + 1 + 2 = 4
Final UCS Answer

- **Optimal Path: S → A → C → G**
- **Minimum Cost: 4**